Soru: 1'den 100'e kadar olan sayıların 2,3,5,7'ye bölünen sayıların elemanı sayısı kaçtır?

Teorem: $|A_1 \cup A_2 \cup \dots A_n| = \sum_i |A_i| - \sum_{i,j} |A_i \cap A_j|$
$+ \sum |A_i \cap A_j \cap A_k|$
$\dots + -1^{n-1} |A_i \cap A_j \cap \dots A_n|$

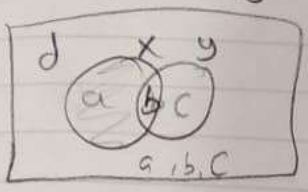
Çözüm: $|A_2 \cup A_3 \cup A_5 A_7| =$

$-(50+33+20+14)$
$+3+2+1+0)$
$-(0)$
$=78$

▶ asalların sayısı: $(100-78)+4-1=25$

$x - y = x \cap \bar{y}$

$x \setminus y = \bar{x} \cap y$



$(a,b \quad b,c)$

2b

$(a,b,) \cap (d,a)$

$\{c\}$

$(c,d) \cap (a,d)$

d

$x \subseteq y$

$A \times B \quad |A| =$

dış değer ↓ iç değeri

$B \times A$

62, 78, 79, 72, 66 ✓

$t^2(A_1 t + A_0)\, et$

$= A_1 t^3 e^t + A_0 e^t . t^2$

oste yuvarla (ceil)

$\lceil 10,3 \rceil = 11$

$\lfloor 10,44 \rfloor = 10$

balta yuvarla (floor)

$\sum_{i=2}^{4} \prod_{j=1}^{3} 3i - 2j = ?$

$(6-2).(6-4).(6-6) = 0$

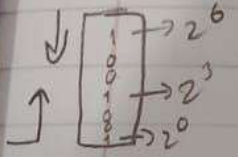
$(9-2).(9-4).(9-6) = 105$

$(12-2).(12-4).(12-6) = 480$

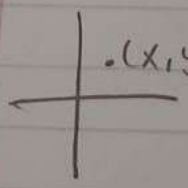
± 585

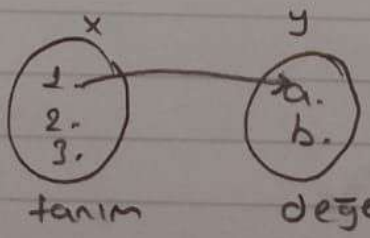
$b^2 a c^3$ bbaccc

$c^3 b^2 a$ cccbba



$y = f(x)$





$(a,b) \subset a \times b$ $\binom{4}{1}\binom{3}{1}$

$\{\text{—}\} \subset (1,a)\ (1,b)\ (2,c)\ (2,b)\ (3,e)\ (3,b)$

$1 \times a \ , 2 \times c$

$A = \{1,2,3,4\}$

1) Yansımalı, $\forall a, [a \propto a]$

2) Yansımasız, $\exists a, [\neg (a \propto c)]$

3) Ters Yansımalı, $\forall a, [\neg (a \propto c)]$

4) Simetri,
$\forall a,b\ [(a=b) \vee (a \propto b) \leftrightarrow (b \propto c)]$

5) Simetrik Olmayan,
$\exists a,b [(a \neq b) \wedge ((a \propto b) \wedge \neg (b \propto a)) \vee$ tersi)

6) Ters Simetrik,
$\forall a,b\ [(a \propto b) \wedge (b \propto c) \neg (a=b)]$

7) Geçişli,
$\forall a,b,c\ [(a \propto b) \wedge (b \propto c) \rightarrow (a \propto c)]$
$(1,2)\quad (2,3) \rightarrow (1,3)$

8) Geçişsiz,
$\exists a,b,c\ [((a \propto b) \wedge (b \propto c)) \wedge \neg (a \propto c)]$

9) Ters Geçişli,
$\forall a,b,c\ [(a \propto b) \wedge (b \propto c)] \rightarrow \neg (a,c)]$

Ör: if $= a=b^2$, $(a,b) \in R$ $A = \{1,2,3,4\}$
$R = \{(1,1), (4,2)\}$